Training Day 11

1. Topics Covered and Brief Explanation

1. Introduction to Supervised Learning

Supervised Learning is a machine learning technique in which a model is trained using labelled datasets. The model learns from input-output examples and uses the learned patterns to predict results for new data.

2. Types of Supervised Learning

Studied two major types:

- Regression: Used to predict continuous numerical values, such as house prices, temperature and student marks.
- Classification: Used to predict categories or classes, such as spam or non-spam emails and positive or negative reviews.

3. Introduction to Unsupervised Learning

Unsupervised Learning is a technique in which a model works with unlabelled data and identifies hidden patterns, similarities and groups without predefined output labels.

4. Clustering

Clustering is an unsupervised learning technique used to group similar data points together based on their characteristics.

For example, customers can be grouped according to their shopping habits and preferences.

5. Difference Between Supervised and Unsupervised Learning

Supervised Learning	**Unsupervised Learning**
Uses labelled data.	Uses unlabelled data.
Has known output values.	Does not have predefined output values.
Used for regression and classification.	Commonly used for clustering.
Learns from input-output examples.	Identifies hidden patterns in data.

2. Tools and Technologies Used

Tool	**Purpose**
Python	Implementing basic ML concepts.
Scikit-learn	Exploring machine learning algorithms.
Pandas	Handling and organizing datasets.
NumPy	Performing numerical operations.

Google Colab Executing Python programs.

3. Practical Learning / Code Example

Example: Simple Classification using Scikit-learn

```
from sklearn.neighbors import KNeighborsClassifier
# Training data
X = [[1], [2], [3], [4], [5], [6]]
y = ["Low", "Low", "Low", "High", "High", "High"]
# Creating the model
model = KNeighborsClassifier(n_neighbors=3)
# Training the model
model.fit(X, y)
# Making prediction
result = model.predict([[5]])
print("Predicted Category:", result[0])
```

Explanation:

- KNeighborsClassifier() creates a classification model.
- n_neighbors=3 means the model considers three nearest neighbours.
- fit() trains the model using labelled data.
- predict() predicts the category for new input data.

4. Learning Outcomes

- Understood the concepts of supervised and unsupervised learning.
- Learned the difference between labelled and unlabelled datasets.
- Gained knowledge of regression and classification techniques.
- Understood the concept of clustering and its applications.
- Gained basic exposure to classification using Scikit-learn.